One-Day Plan: Passkey Memory Vault-Lite on Monad

Prepared 9 October 2026 for the Metropolis hackathon. With only about one day to build, the scope is cut to one demo flow. Verify exact bounty requirements before submitting.

Do now (15 minutes)

Create the project in the dashboard and select **Track 4** plus the bounties below. Until you do this you can't submit. Also fund a testnet wallet from the faucet.

What to build (one flow only)

1. Log in with Mera (passkey).
2. Paste or upload one text document. It is encrypted in the browser with a Mera-derived key and stored in Supabase.
3. A **Grant** contract on Monad: `grant(agent, docId, expiry)`, `revoke(...)`, and events for each. About 40 lines of Solidity.
4. Ask a question. The app checks the grant, decrypts only the relevant chunks, sends them to Kimi (or Qwen), and shows the answer with the source chunks.
5. Click revoke, then ask again. The app refuses and shows the on-chain event.

What to cut

Envio, multi-document support, fancy RAG (embed nothing, just send the doc in chunks), roles and teams, and polish beyond one clean screen. Alchemy is only worth it if you can simply use its RPC URL for Monad (check that it's supported); don't build around it.

Bounties to target

Track 4, both Mera bounties, and one of Kimi or Qwen. That is still a realistic stack. Don't promise Envio in your submission if you didn't build it.

Hour budget (about 12-14 hours of work)

1. **Hours 0-2, gate:** Mera login, key derivation, encrypt and decrypt round trip.
2. **Hours 2-4:** Grant contract deployed on Monad testnet, with revoke.

3. **Hours 4-7:** Gateway that checks the grant, decrypts chunks, calls the LLM and shows the answer.
4. **Hours 7-9:** UI wiring, plus the revoke-then-reject path.
5. **Hours 9-11:** Deploy to Vercel and test end to end twice.
6. **Hours 11-13:** Demo video (about 2 minutes), README and submission form.

The gate matters more now

If Mera's key derivation isn't working by hour 2, don't keep fighting it. Fall back to a key derived another way (for example a random key held by the user). You then lose the "One Passkey, Many Keys" bounty, but you still have a working demo for Track 4 and Best Mera UX. A working app with fewer bounties beats a broken one.

Be honest in the demo

Say that revocation stops future access but can't un-send data an agent already received. Judges respond well to a stated limit.

Next step

Decide whether to have the Solidity contract and the encrypt/decrypt helper written first, so coding can start immediately.